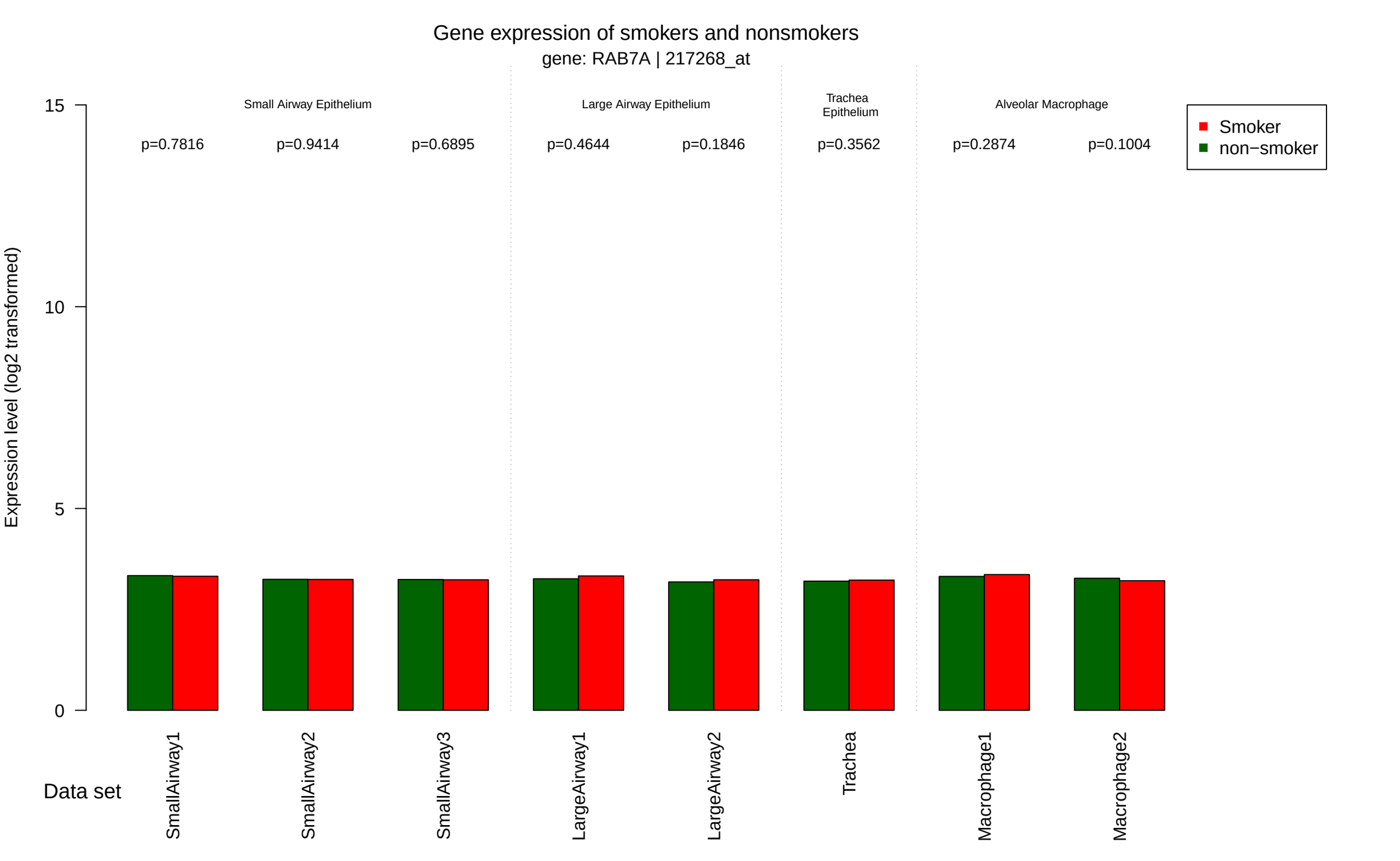
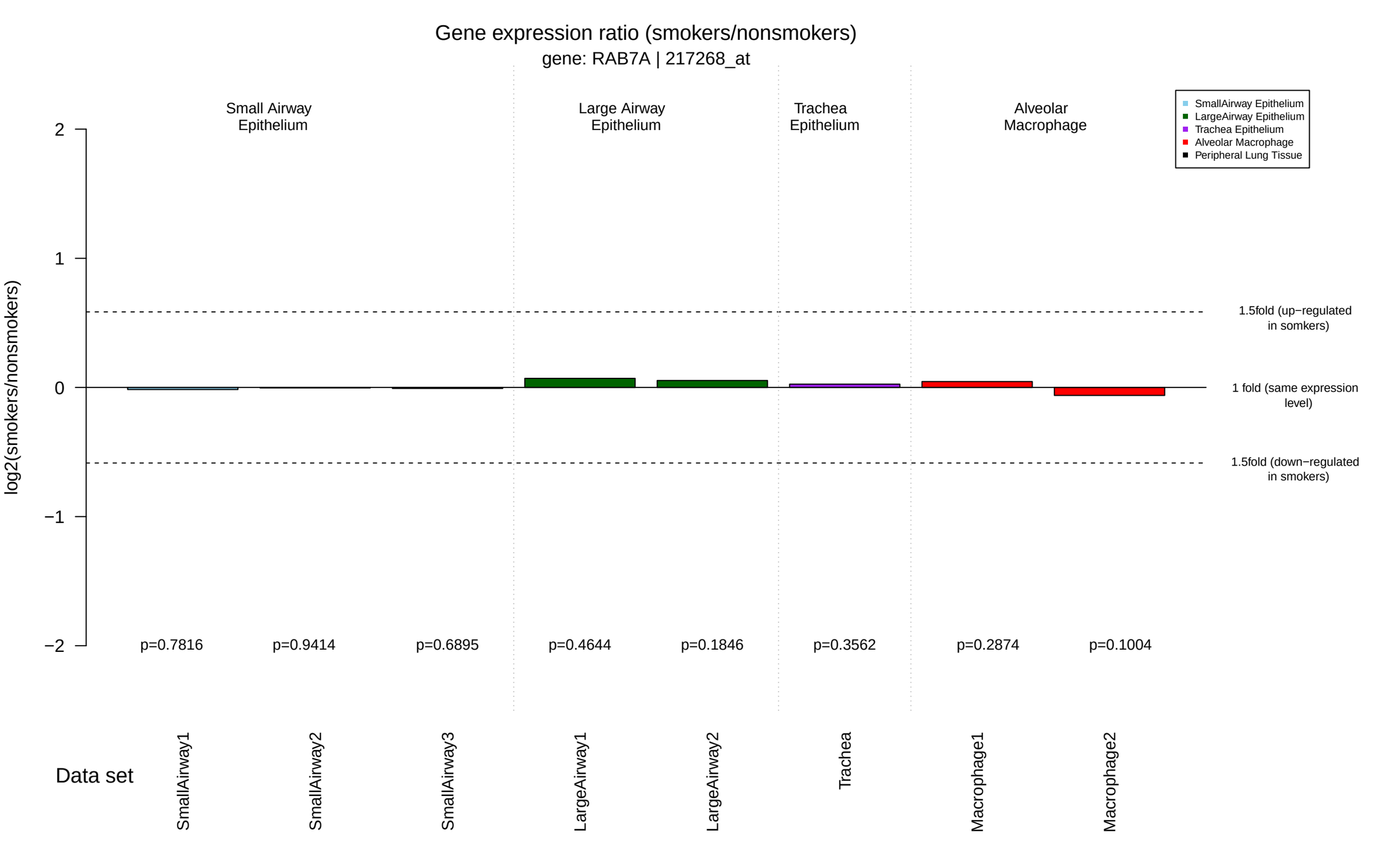
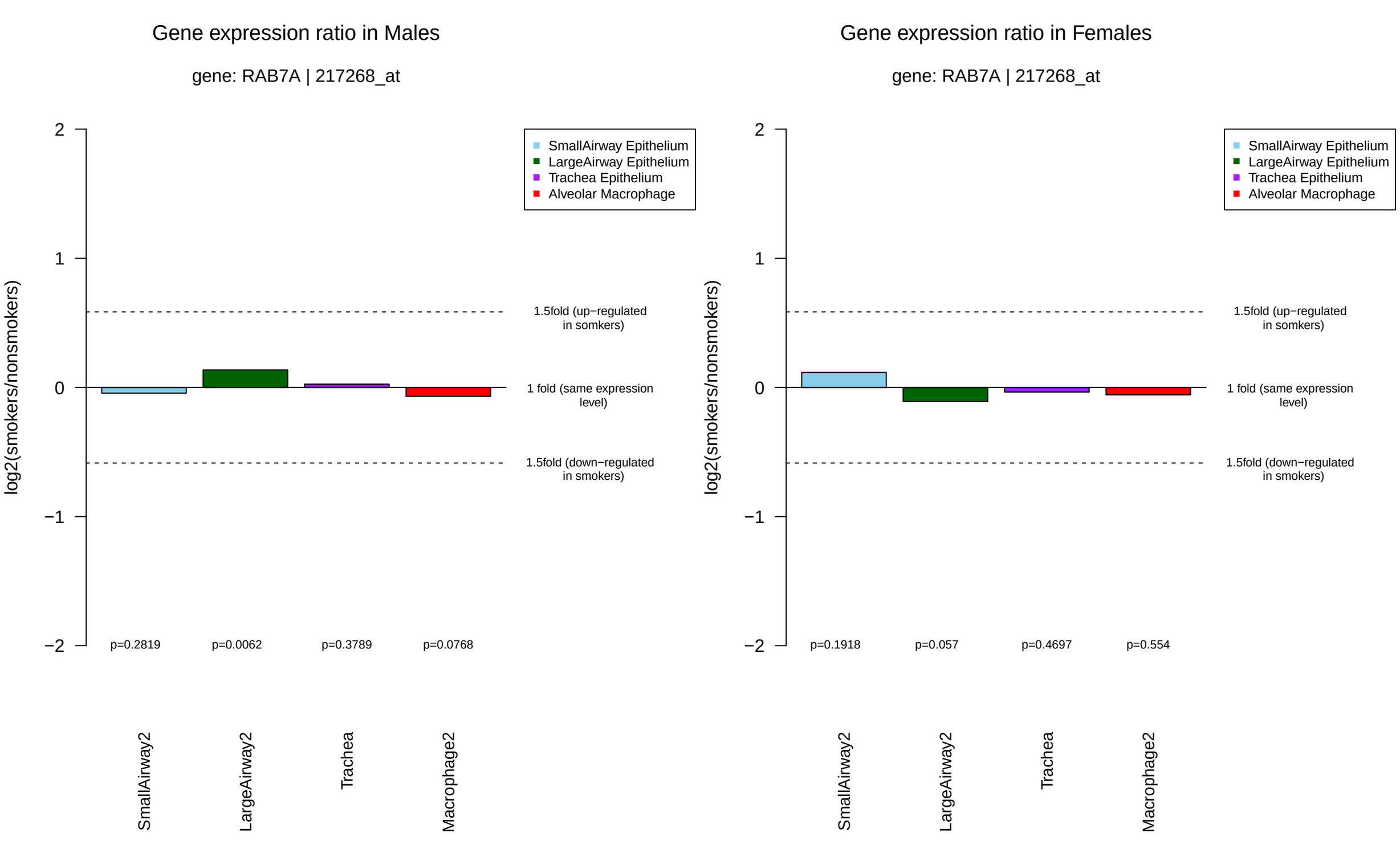
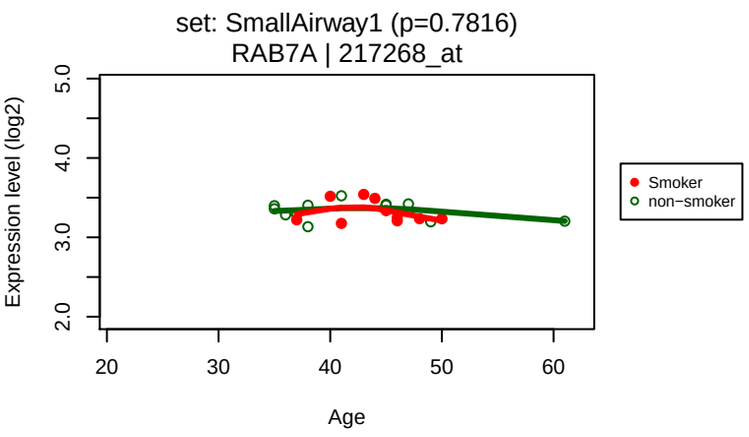
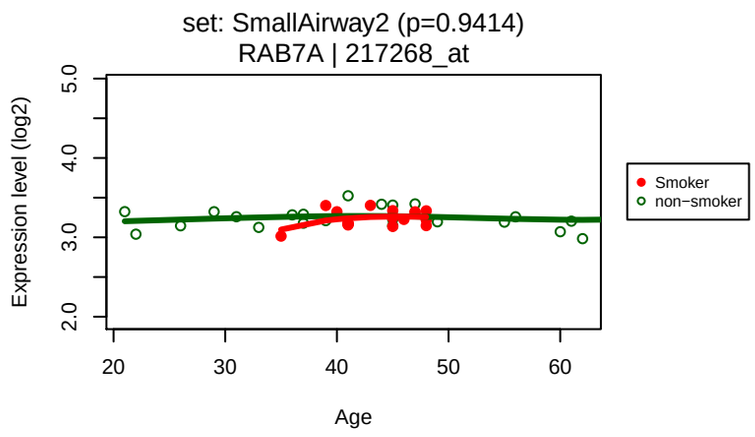
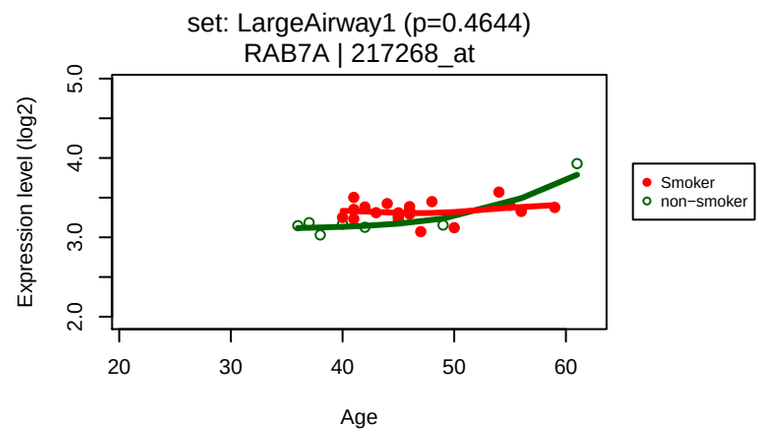

Gene expression of smokers and nonsmokers
gene: RAB7A | 217268_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7816
p=0.9414
p=0.6895
p=0.4644
p=0.1846
p=0.3562
p=0.2874
p=0.1004
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

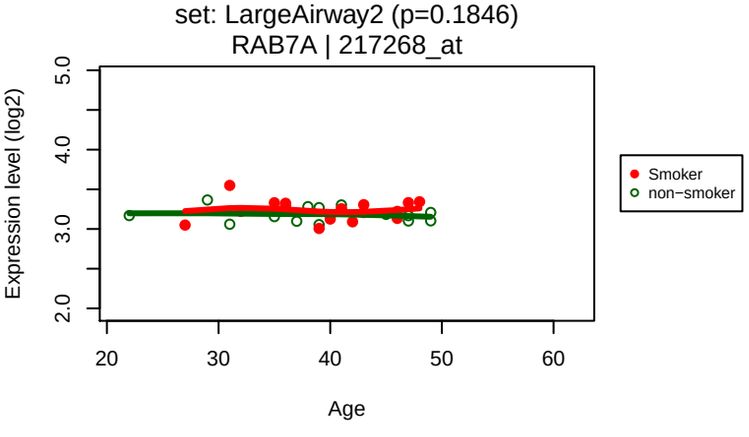

Gene expression ratio (smokers/nonsmokers)
gene: RAB7A | 217268_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7816
p=0.9414
p=0.6895
p=0.4644
p=0.1846
p=0.3562
p=0.2874
p=0.1004
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

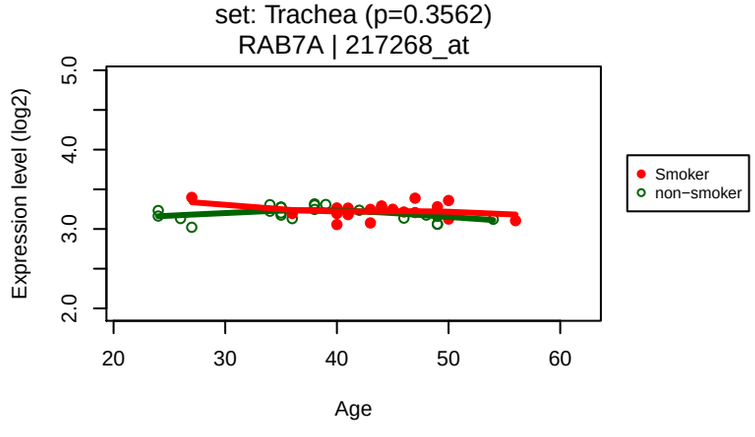

Gene expression ratio in Males
gene: RAB7A | 217268_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2819
p=0.0062
p=0.3789
p=0.0768
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB7A | 217268_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1918
p=0.057
p=0.4697
p=0.554
SmallAirway2
LargeAirway2
Trachea
Macrophage2

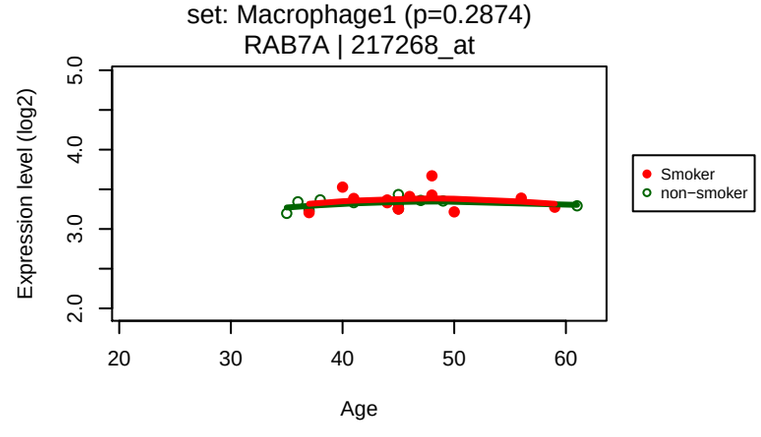

set: SmallAirway1 (p=0.7816)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker

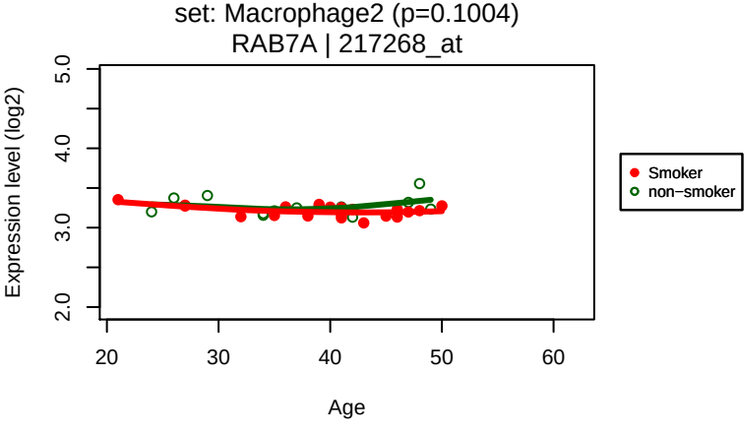

set: SmallAirway2 (p=0.9414)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker

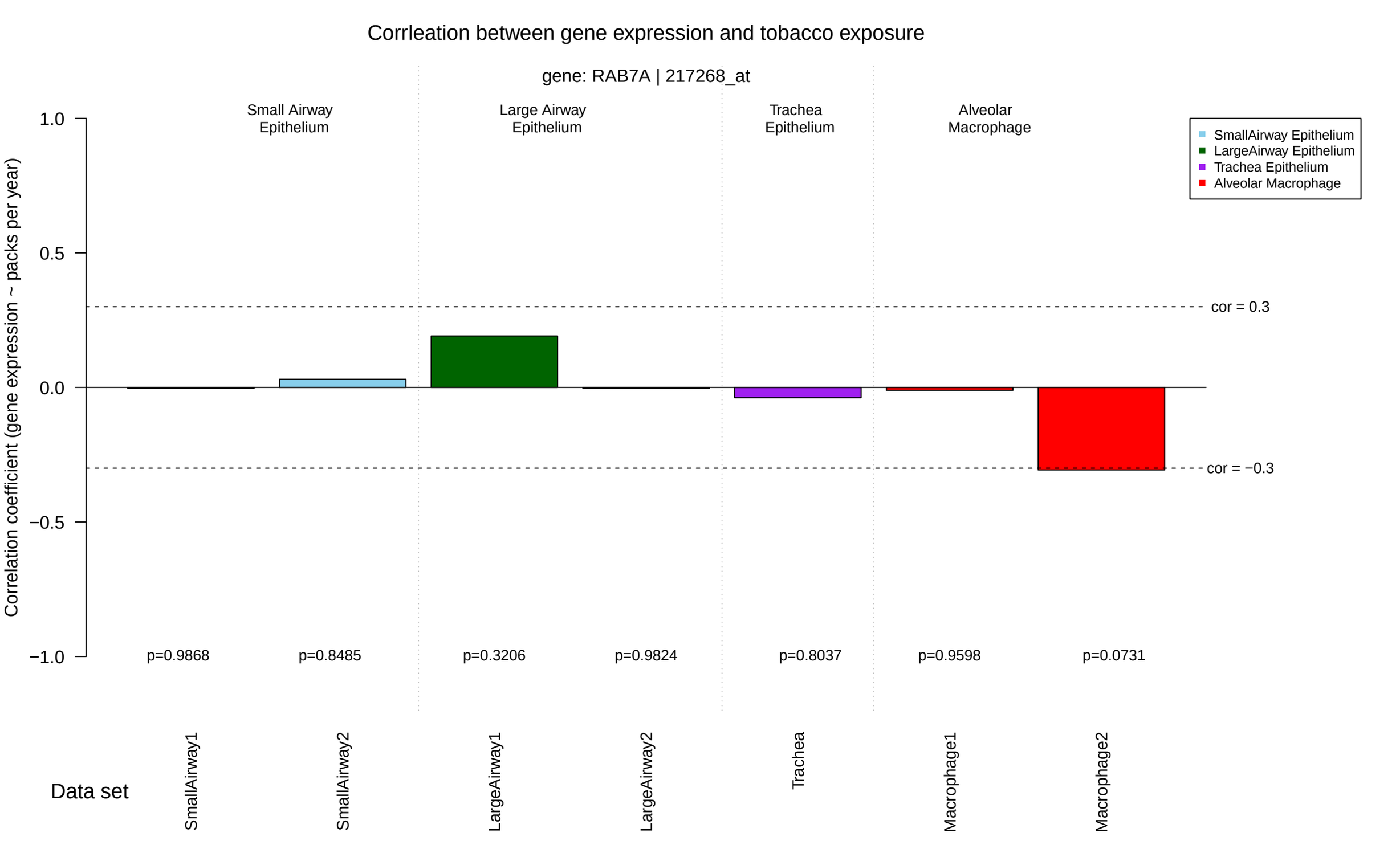

set: LargeAirway1 (p=0.4644)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1846)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3562)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2874)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1004)
RAB7A | 217268_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB7A | 217268_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9868
p=0.8485
p=0.3206
p=0.9824
p=0.8037
p=0.9598
p=0.0731
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2